

**YEAR 9 SCIENCE
PHYSICAL SCIENCES
INVESTIGATION - THE SPEED OF LIGHT**

Aim: To determine the speed of light.

Equipment:

- Microwave oven
- Family block of chocolate
- Ruler
- Small ceramic bowl

Method:

1. Raise the turntable in the microwave by placing an upturned ceramic bowl over the rotator. **The turntable must not turn.**
2. Unwrap the chocolate bar and place it in the centre of the turntable with the **smooth side facing upwards**.
3. Start the microwave and watch the chocolate until evidence of melting occurs. This may be as the chocolate starts to discolour or may go "shiny".
4. Stop the microwave and allow the chocolate to cool.
5. Use a ruler to measure the distance between the "hot spots" that show on the chocolate. Record them in the table below and find the average.
6. Record the frequency of the microwave - it is on the plate at the back of the appliance. It is usually in gigahertz (GHz).

Background:

The speed of any wave is given by:

$$\text{speed} = \text{frequency} \times \text{wavelength}$$

The frequency will need to be converted to Hertz (Hz).

The wavelength needs to be in metres (m).

This will give the answer in metres/second (m/s).

The microwaves inside the oven bounce around and form what are called "standing waves". This means they have distinct points where the energy adds together, causing "hot spots".

The distance between adjacent "hot spots" is half of a wavelength.

Results:

Distance between "hot spots" (cm)

Average distance between "hot spots" = _____ cm

= _____ m

Wavelength = 2 x average distance = _____ m

Frequency: _____ GHz (multiply by 1,000,000,000)

= _____ Hz

Speed = frequency x wavelength

= _____ x _____

= _____ m/s

Conclusion:

The speed of light is _____ .